

Introduction to Contract Theory and Auctions  
Lecture 3  
Choice under Uncertainty & Expected Utility Theory

Prof. Alejandro Melo-Ponce



NAZARBAYEV  
UNIVERSITY

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## Decision making under uncertainty

- When thinking about decision making, we usually distinguish between action vs consequences.
- Action  $\rightarrow$  consequence.
- In the standard choice model that you have seen in your previous courses, a decision maker has preferences over the set of consequences and she is meant to choose a feasible action that leads to the most desired consequence.
- However, in that previous discussion, it was not necessary to distinguish between actions vs consequences, because the link between action  $\rightarrow$  consequence was **deterministic**.
- We will now consider a decision maker in an environment in which this link is not deterministic but **stochastic**.
  - Example: raffle ticket that yields a car with probability 0.001 and nothing otherwise.
  - Example: Ski trip in which you will experience snowstorm with probability 0.3 and good weather with probability 0.7.
- We will model these randomness using the concept of a **lottery**.

## Goals for this lecture

- In your past classes you modeled a decision maker (DM) making her choices out from a choice set  $X$  in which this set could be
  - Finite, i.e.,  $X = \{a, b, c\}$ .
  - Infinite but countable, i.e.,  $X = \{a_i, i \in \mathbb{N}\}$ .
  - Infinite and uncountable, i.e.  $X = \mathbb{R}_+^n$ .
- Then you studied the DM's preferences over  $X$  and analyze conditions under which her preferences could be represented by a utility function  $u : X \rightarrow \mathbb{R}$ .
- You used  $u$  as a tool to study the DM's behavior and predict her choices.
- What we will do in this lecture is to develop a model in which the choice set  $X$  incorporates the notion of randomness that we discussed before, study the kind of preferences over this set that the decision maker might have, and then study the conditions under which we can guarantee the existence of a utility function representation **with a special form** that represents the DM's preferences.

- The starting point is a set  $Z$  of objects called prizes.
  - $Z$  can be finite or infinite, although infinite set of prizes introduces some technical difficulties.
- A **lottery** will be defined over the set  $Z$  of prizes, and specifies the probability with which each prize is realized.
- For the moment, I will focus on sets of prizes  $Z$  that are finite.

## Definition 3.1

Let  $Z$  be a finite<sup>a</sup> set of prizes. A lottery over  $Z$  is a function  $p : Z \rightarrow \mathbb{R}$  that assigns a positive number (probability)  $p(z)$  to a finite number of members of  $Z$  and 0 to all other members, with  $\sum_{z \in Z} p(z) = 1$ . The **support** of the lottery  $p$ , denoted  $\text{supp}(p)$ , is the set  $\{z \in Z : p(z) > 0\}$ . We denote the set of all lotteries over  $Z$  as  $L(Z)$ . We denote the lottery that yields prize  $z$  with probability 1 as  $[z]$ , and the lottery that yields prize  $z_k$  with probability  $\alpha_k$  for  $k = 1, \dots, K$  by  $\alpha_1 \cdot z_1 \oplus \alpha_2 \cdot z_2 \oplus \dots \oplus \alpha_K \cdot z_K$ .

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<sup>a</sup>The extension to infinite sets requires a little bit of background in measure theory in order to be defined properly.

## The space of lotteries $L(Z)$

- Note that although we are considering the set  $Z$  to be finite, nevertheless,  $L(Z)$  is an infinite set, since it is the set of all probability mass functions over  $Z$ .
- In fact, when  $Z$  is finite and its cardinality is  $|Z| = n$ , we can identify  $L(Z)$  with a subset of  $n$ -dimensional Euclidean space.
- $L(Z)$  is the standard simplex, formed with the  $n$  standard unit vectors, or

$$L(Z) = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1, x_i \geq 0, i = 1, \dots, n\}$$

- Notice that  $L(Z)$  is a  $n - 1$  dimensional object.
- The standard simplex is a **convex** set, it is actually a convex polytope; it is the convex hull of the  $n$  standard unit vectors.
- We are interested in **preference relations** over  $L(Z)$ , since preference relations may be a new concept for you, I will make a brief detour.

## Preferences over lotteries

Consider the following examples of “sound” preferences over  $L(Z)$

### Example 3.2 (A pessimist)

Suppose that the DM has a strict preference relation  $\succsim^*$  over the set  $Z$  of prizes. The DM evaluates lotteries by the *worst possible case*. That is, she prefers the lottery  $p \in L(Z)$  to the lottery  $q \in L(Z)$  if she prefers the worst prize that occurs with positive probability in  $p$  to the worst prize that occurs with positive probability in  $q$ . Formally, define  $w(p)$  to be a prize in  $\text{supp}(p)$  such that  $y \succsim^* w(p)$  for every  $y \in \text{supp}(p)$ . Then the DM's preference relation over  $\succsim$  over  $L(Z)$  is defined by  $p \succsim q \Leftrightarrow w(p) \succsim^* w(q)$ .

### Example 3.3 (Good and Bad)

The DM divides the set  $Z$  of prizes into two subsets, good and bad. For any lottery  $p \in L(Z)$ , let  $G(p) = \sum_{z \in \text{good}} p(z)$  be the total probability that a good prize occurs. The DM prefers the lottery  $p \in L(Z)$  to the lottery  $q \in L(Z)$  if the probability of a prize in good occurring is at least as high for  $p$  as it is for  $q$ . Formally,  $p \succsim q \Leftrightarrow G(p) \geq G(q)$ .

## Example 3.4 (Minimizing options)

A DM wants the number of prizes that might be realized (the number of prizes in the support of the lottery) to be as small as possible. Formally,  $p \succsim q \Leftrightarrow |\text{supp}(p)| \leq |\text{supp}(q)|$ . This preference relation makes sense for an individual who does not care about the realization of the lottery but wants to be as prepared as possible for all possible outcomes.

## Example 3.5 (Expected Utility)

A number  $v(z)$  is attached to each prize, and a lottery  $p$  is evaluated according to its expected  $v$ , that is according to  $\sum_{z \in Z} p(z)v(z)$ . Thus,

$$p \succsim q \Leftrightarrow U(p) = \sum_{z \in Z} p(z)v(z) \geq U(q) = \sum_{z \in Z} q(z)v(z).$$

- Preference relations over lotteries can take an unlimited number of other forms. For example, note that the above examples can be combined in various ways to obtain even more examples.
- This richness of examples calls for the classification of preference relations over lotteries and the study of properties that these relations satisfy.
- The methodology we follow is to formally state general principles (axioms) that may apply to preferences.
- Each axiom carries with it a consistency requirement or involves a procedural aspect of the decision making.

- The axiomatization that we will study uses the **independence** and **continuity axioms**.
- To state the independence axiom, we first need the concept of a **compound lottery**.

## Definition 3.6 (Compound lottery)

Let  $Z$  be a set of prizes, and let  $p_1, \dots, p_K$  be lotteries in  $L(Z)$ , and let  $\alpha_1, \dots, \alpha_K$  be nonnegative numbers that add up to 1. The **compound lottery**  $\alpha_1 \cdot p_1 \oplus \dots \oplus \alpha_K \cdot p_K$  is the lottery that yields each prize  $z \in Z$  with probability  $\sum_{k=1}^K \alpha_k p_k(z)$ .

- Notation  $\alpha_1 \cdot p_1 \oplus \dots \oplus \alpha_K \cdot p_K = \oplus_{k=1}^K \alpha_k \cdot p_k$ .
- You can think of the compound lottery  $\oplus_{k=1}^K \alpha_k \cdot p_k$  as occurring in two stages:
  - Stage 1. It is randomly determined which lotteries  $p_1, \dots, p_K$  is realized;  $\alpha_k$  is the probability that  $p_k$  is realized.
  - Stage 2. The prize received is randomly drawn from the lottery determined in stage 1.

## Definition 3.7 (Independence (I))

For any  $p, q, r \in L(Z)$  and any  $\alpha \in (0, 1)$ ,

$$p \succsim q \Leftrightarrow \alpha \cdot p \oplus (1 - \alpha) \cdot r \succsim \alpha \cdot q \oplus (1 - \alpha) \cdot r.$$

The following property follows from  $I$ .

## Definition 3.8 ( $I^*$ )

For every  $(p_k)_{k=1,\dots,K}$ , a vector of lotteries,  $q_{k^*}$  a lottery, and  $(\alpha_k)_{k=1,\dots,K}$  an array of nonnegative numbers such that  $\alpha_{k^*} > 0$  and  $\sum_k \alpha_k = 1$ ,

$$\bigoplus_{k=1}^K \alpha_k \cdot p_k \succsim \bigoplus_{k=1}^K \alpha_k \cdot q_k \text{ when } p_k = q_k \text{ for all } k \text{ but } k^* \Leftrightarrow p_{k^*} \succsim q_{k^*}$$

# What happens if we don't assume the independence axiom?

- To understand the relevance of the independence axiom, it is worth considering the difficulties that arise when we drop it.
- Suppose that  $[x] \succ [y]$  but  $0.5[y] + 0.5[z] \succ 0.5[x] + 0.5[z]$ .
- Suppose  $w$  is another prize such that  $0.5[y] + 0.5[z] \succ [w] \succ 0.5[x] + 0.5[z]$ .
- The DM must decide whether to take  $[w]$  or not.
  - If he doesn't then a coin is tossed.
    - If Heads, he gets  $[z]$ .
    - If Tails, he gets to choose between  $[x]$  and  $[y]$ .
- Three possibilities:
  - ① Take  $[w]$
  - ② Reject  $[w]$ , choose  $[x]$  if Tails  $\Leftrightarrow 0.5[z] + 0.5[x]$
  - ③ Reject  $[w]$ , choose  $[y]$  if Tails  $\Leftrightarrow 0.5[z] + 0.5[y]$
- (3) is better than both (2) and (1), so DM should go for (3)
- However, by refusing  $w$ , if the coin comes out Tails, he faces a choice between two sure lotteries ( $[x]$  vs  $[y]$ ).
- His preferences dictate that he should choose  $[x]$  if the coin comes Tails.
- Thus, if he rejects  $[w]$  he should go for (2) ending up with  $0.5[z] + 0.5[x]$  but this is worse than  $[w]$ , contradiction!
- In the absence of the axiom we must specify if a rational DM can commit to a strategy (3) that he would later regret.
- If the DM can't make such commitment, we must specify if he can foresee their inconsistency (1) or if they can't foresee (2).

We can derive other useful implications from the independence axiom (I). The following is called **monotonicity**.

## Lemma 3.9 (Monotonicity (M))

Let  $\succsim$  be a preference over  $L(Z)$  satisfying I. Let  $x, y \in Z$  such that  $[x] \succ [y]$  and  $1 \geq \alpha > \beta \geq 0$ . Then,

$$\alpha x \oplus (1 - \alpha)y \succ \beta x \oplus (1 - \beta)y$$

## Proof.

If either  $\alpha = 1$  or  $\beta = 0$ , the claim follows directly from I. So suppose that  $1 > \alpha > \beta > 0$ . Then, by I  $\alpha \cdot x \oplus (1 - \alpha) \cdot y \succ [y]$ . Using I again we get  $\alpha \cdot x \oplus (1 - \alpha) \cdot y \succ (\beta/\alpha) \cdot (\alpha \cdot x \oplus (1 - \alpha) \cdot y) \oplus (1 - \beta/\alpha)[y] = \beta x \oplus (1 - \beta)y$ . □

# The Continuity Axiom

- We will employ the same continuity assumption as in the last lecture.
- In this environment, continuity means that the preferences are not overly sensitive to small changes in probabilities.

## Definition 3.10 (Continuity (C))

If  $p \succ q$ , then there are neighborhoods<sup>a</sup>,  $B(p)$  of  $p$  and  $B(q)$  of  $q$  such that:

$$\text{for all } p' \in B(p) \text{ and } q' \in B(q), p' \succ q'.$$

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<sup>a</sup>Remember that since we assumed that  $Z$  is finite,  $L(Z)$  is just a subset of  $\mathbb{R}^{|Z|}$ .

The following is an alternative definition of continuity:

## Definition 3.11 (C\*)

If  $p \succ q \succ r$ , then there exists  $\alpha \in (0, 1)$  such that

$$q \sim \alpha \cdot p \oplus (1 - \alpha) \cdot r.$$

## Expected Utility is Continuous and Independent

### Definition 3.12 (Expected Utility)

For any set of prizes  $Z$ , a preference relation  $\succsim$  on  $L(Z)$  is consistent with expected utility if there exists a function  $v: Z \rightarrow \mathbb{R}$  such that  $\succsim$  is represented by the utility function  $U$  defined by  $U(p) = \sum_{z \in Z} p(z)v(z)$  for each  $p \in L(Z)$ . The function  $v$  is called the **Bernoulli function** for the representation.

### Proposition 3.13 (Expected utility is continuous and independent)

*A preference relation  $\succsim$  on  $L(Z)$  that is consistent with expected utility, satisfies the continuity and independence axioms.*

## Proof of proposition 3.13

### Proof.

Let  $Z$  be a set of prizes, let  $\succsim$  be a preference relation over  $L(Z)$ , and let  $v: Z \rightarrow \mathbb{R}$  be a function such that the function  $U$  defined by  $U(p) = \sum_{z \in Z} p(z)v(z)$  for each  $p \in L(Z)$  represents  $\succsim$ .

**Continuity (C\*)** Let  $p, q, r \in L(Z)$  satisfy  $p \succ q \succ r$ . Since  $U$  is a utility representation, then  $U(p) > U(q) > U(r)$ . Let  $\alpha$  satisfy  $\alpha U(p) + (1 - \alpha)U(r) = U(q)$ , i.e.,  $0 < \alpha = (U(q) - U(r))/(U(p) - U(r)) < 1$ . The previous sentence implies that  $\alpha \cdot p \oplus (1 - \alpha) \cdot r \sim q$ .

**Independence** Note that the function  $U(p)$  is linear:

$$\begin{aligned} U(\oplus_{k=1}^K \alpha_k \cdot p_k) &= \sum_{z \in Z} [\oplus_{k=1}^K \alpha_k \cdot p_k](z) v(z) = \sum_{z \in Z} \left[ \sum_{k=1}^K \alpha_k p_k(z) \right] v(z) \\ &= \sum_{k=1}^K \alpha_k \left[ \sum_{z \in Z} p_k(z) v(z) \right] = \sum_{k=1}^K \alpha_k U(p_k). \end{aligned}$$

It follows that any such preference relation satisfies I.

## The expected utility theorem of von Neumann and Morgenstern

- By Debreu's theorem from last lecture, any continuous preference relation  $\succsim$  on  $L(Z)$ , there is a utility representation  $U: L(Z) \rightarrow \mathbb{R}$  continuous in the probabilities.
- However, Debreu's theorem puts no restriction on the shape of such a utility representation.
- It turns out that the von Neumann and Morgenstern axiomatization, i.e., assuming axioms I and C, yields an **expected utility representation**.

### Theorem 3.14 (vNM)

Let  $\succsim$  be a preference relation over  $L(Z)$  satisfying axioms I and C. Then, there are numbers  $(v(z))_{z \in Z}$  such that:

$$p \succsim q \Leftrightarrow U(p) = \sum_{z \in Z} p(z)v(z) \geq U(q) = \sum_{z \in Z} q(z)v(z).$$

## Proof of the vNM theorem.

### Proof.

Let  $[M]$  and  $[m]$  be a best and a worst lotteries in  $L(Z)$  (How do we know they exist?).

Consider first the case in which  $[M] \sim [m]$ . It follows from  $I^*$  that  $p \sim m$  for any  $p$  and thus  $p \sim q$  for all  $p, q \in L(Z)$ . Thus, any constant utility function represents  $\succsim$ . Choosing  $v(z) = 0$  for all  $z \in Z$ , we have  $\sum_{z \in Z} p(z)v(z) = 0$  for all  $p \in L(Z)$ .

Now consider the case in which  $[M] \succ [m]$ . By  $C^*$  and  $M$ , there is a unique number  $v(z) \in [0, 1]$  such that  $v(z)M \oplus (1 - v(z))m \sim [z]$  for every prize  $z \in Z$ . In particular, we can put  $v(M) = 1$  and  $v(m) = 0$ . By  $I^*$  we obtain that:

$$p \sim \left( \sum_{z \in Z} p(z)v(z) \right) M \oplus \left( 1 - \sum_{z \in Z} p(z)v(z) \right) m.$$

Once again, by monotonicity,  $p \succsim q$  if and only if  $\sum_{z \in Z} p(z)v(z) \geq \sum_{z \in Z} q(z)v(z)$ . □

- The vNM utilities are unique up to positive affine transformations.
- They are not invariant to arbitrary monotonic transformations.
- Let  $\succsim$  be a preference relation over  $L(Z)$  and let  $v(z)$  be the VNM utilities representing the preference relation.
- By defining  $w(z) = \alpha v(z) + \beta$  for all  $z \in Z$  and for  $\alpha > 0$  and some  $\beta$ , the utility function  $W(p) = \sum_{z \in Z} p(z)w(z)$  also represents  $\succsim$ .

## Uniqueness of the vNM utilities

- We can also show that the converse is true, i.e., that if  $W(p) = \sum_{z \in Z} p(z)w(z)$  represents  $\succsim$  as well, then it must be a positive affine transformation of  $v$ .
- To see this, let  $\alpha > 0$  and  $\beta$  solve the following system of equations:

$$w(M) = \alpha v(M) + \beta \quad w(m) = \alpha v(m) + \beta,$$

where  $M$  and  $m$  are the best and worst prizes, as defined the last slide. (What ensures the existence of  $\alpha > 0$  and  $\beta$ ?)

- For any  $z \in Z$ , there is  $p = [0, 1]$  such that  $[z] \sim pM \oplus (1 - p)m$ , and therefore, it must be that

$$\begin{aligned} w(z) &= pw(M) + (1 - p)w(m) \\ &= p(\alpha v(M) + \beta) + (1 - p)(\alpha v(m) + \beta) \\ &= p[\alpha v(M) + (1 - p)v(m)] + \beta \\ &= \alpha v(z) + \beta. \end{aligned}$$

- An implicit assumption in the above formulation of the lotteries framework is that the decision maker does not care about the nature of the random factors but only about the distribution of consequences.
- To appreciate this point, consider a case in which the probability of rain is  $1/2$  and  $Z = \{z_1, z_2\}$ , where  $z_1$  = having an umbrella and  $z_2$  = not having an umbrella.
- A lottery in which you have  $z_1$  if it is raining and  $z_2$  if it is not should not be considered equivalent to the lottery in which you have  $z_1$  if it is not raining and  $z_2$  if it is.
- Thus, we must be careful to apply the model studied in this lecture in context where the attitude towards the consequence depends on the event realized in each possible contingency.

- Regarding the plausibility of the vNM axiomatization as a model of human behavior, there is a large literature in both psychology and economics providing counterarguments. (You can take the Behavioral Economics class to see more about this.)
- In particular, it has been shown that when dealing with uncertainty, individual preferences are highly sensitive to the **framing** of the alternatives:
- What would you choose first between:
  - ①  $L_1 = 0.25[3,000] + \oplus 0.75[0]$
  - ②  $L_2 = 0.2[4,000] \oplus 0.8[0]$
- and then choose between:
  - ①  $L_3 = [3,000]$
  - ②  $L_4 = 0.8[4,000] \oplus 0.2[0]$

# The Allais paradox

- Note that  $L_1 = 0.25L_3 \oplus 0.75[0]$  and  $L_2 = 0.25L_4 \oplus 0.75[0]$ .
- Axiom I requires that the preference between  $L_1$  and  $L_2$  be in the same direction as that between  $L_3$  and  $L_4$ .
- However, experiments show that:
  - the majority of people say  $L_1 \prec L_2$ ,
  - but an even larger majority say  $L_3 \succ L_4$ .
- What is the difference? We could argue that  $L_3$  and  $L_4$  are presented as simple lotteries, whereas  $L_1$  and  $L_2$  are presented as compound lotteries.
- This is a version of the Allais paradox.

## More discussion points

- The independence axiom also does not take into account any notion of “fairness” of the random process.
- Suppose you were a kind uncle and had a niece and a nephew
- You only have one gift.
- Your option is to choose a lottery  $L(p)$  where  $p$  is the probability that awards the gift to your niece.
- You do not favor your niece over your nephew.
- Independence implies that you are indifferent between any  $p$ .
- Is this a suitable model of your behavior?

All the previous issues in the discussion are important, but we will carry all of our analysis within the context of the vNM axiomatization.